

Tensor Network Solver for the 2D Convecting Taylor-Green Vortex

Quantum-Inspired Finite-Volume Compression
and Measured Scaling Analysis

2026 Global Quantum + AI Challenge
Airbus: Quantum Solvers - Enhancing Predictive
Aerodynamic Modeling Capabilities

Submission track: Tensor Network solver within a Finite Volume framework
(explicitly permitted by the challenge statement)

Experimental research output - not a validated engineering deliverable.

Phase 1 submission - 4 July 2026

1. Executive Summary

We present a verified 2nd-order finite-volume baseline for the prescribed 2D Convecting Taylor-Green Vortex (TGV), then report measured evidence that octonion-associator-guided tensor compression provides real advantages over naive alternatives - including a measured 1010x trajectory-storage reduction at machine-precision error on a 256^2 grid - with an honest account of what did and did not survive our own control experiments, and a concrete qubit-count mapping to future fault-tolerant quantum hardware.

Pursuing the explicitly allowed 'Tensor Network solver within a Finite Volume framework' track, we demonstrate a classical tensor-network memory advantage today and map the pathway to logarithmic quantum qubit scaling for future fault-tolerant hardware. No quantum computer is used, as the challenge track permits.

Every number in this report regenerates from a documented script on a standard Windows CPU. Nothing is estimated where it could be measured; where we project, we state the measurement basis explicitly.

2. Challenge Requirements and Where They Are Met

Challenge requirement	Delivered evidence
Working 2D Convecting TGV solver	Verified FV solver, order 1.99, 48/48 tests
Scaling: time / memory / error vs Re	Measured Re = 10 to 2000 (metrics CSV)
Re = 10, 100 and beyond	Re = 10, 100, 250, 500, 1000, 2000
Comparison vs classical baseline	1010x memory (measured), 1.73x rank advantage
Tensor Network within FV framework	Tucker space-time compression (code supplied)

3. The Solver Is Verified (Foundation for Everything Else)

The baseline is a cell-centered finite-volume solver on a periodic $[0, 2\pi]^2$ domain: face-based convective and diffusive fluxes, RK2 time stepping with CFL control, and FFT (spectral) pressure projection enforcing incompressibility. The benchmark truth is the exact analytical convecting TGV solution, so no external CFD data is required.

Property	Measured value
Grid-convergence order (Re=100, 32^2 to 128^2)	1.99 (design order 2)
L2 velocity error at 32^2 / 64^2 / 128^2	0.00602 / 0.00178 / 0.00045
Divergence L2 (FFT projection)	$\sim 1e-7$
Stable Reynolds range	Re = 10 - 2000 (20x mandatory Re=100)
Automated tests	48 / 48 pass

This verification baseline ensures that every compression or steering effect below is measured against a credible classical solver, not a broken one.

4. Tensor-Network Compression: Measured Through 256^2

The full space-time (time, x, y) solution tensor is stored in Tucker-decomposed form (TensorLy), with per-mode ranks selected automatically from octonion-associator torsion statistics. This implements the Tensor Network within Finite Volume pathway named in the challenge statement.

Grid	Dense	Tucker (guided)	Ratio	Error	Status
64^2	2.1 MB	0.27 MB	7.8x	$1.1e-15$	measured
128^2	32.8 MB	0.32 MB	102x	$1.2e-15$	measured
256^2	523 MB	0.52 MB	1010x	$1.4e-15$	measured

512 ²	4.1 GB	0.86 MB	4887x	-	projected
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Mechanism (measured, not assumed): guided ranks saturate at (14, 29, 29) from 128² onward because rank tracks flow complexity, not resolution. We verified this by decomposing the real 256² trajectory at the smaller grids' ranks and recovering identical machine-precision error. Dense storage grows as $O(n^3)$; compressed storage barely grows. The 512² projection therefore rests on measured rank stability, not extrapolated rank stability.

Rank-selection policy (for reproducibility): ranks come from `associator_guided_ranks` with a grid-independent cap (min 4, max 32). An earlier internal draft used a grid-dependent cap, which inflated ranks with no error benefit; we corrected the policy and re-measured everything above.

Caveat, stated up front: the TGV is spectrally narrow and favorable to low-rank methods. Industrially relevant multi-mode flows would compress less. That is exactly why we ran the perturbed-flow experiment in Section 5.

5. The Associator's Contribution Grows with Flow Complexity

At matched reconstruction error, associator-guided asymmetric rank allocation is compared head-to-head against the best uniform fixed rank on the same trajectories ($Re=100$, $n_x=64$, $t_{\text{final}}=0.5$). Perturbations are divergence-free multi-mode streamfunction fields - a physically meaningful harder flow.

Flow	Guided ranks	Compression	Error	Advantage
Smooth TGV	(15, 30, 30)	7.8x	1.1e-15	1.22x
Perturbed 10%	(13, 26, 26)	11.1x	8.0e-8	1.73x
Perturbed 30%	(11, 22, 22)	16.4x	5.3e-6	1.67x

The guided allocation matters more as the flow gains structure - the direction that matters for real aerodynamic flows. Ranks also adapt automatically to Reynolds number (spatial rank 19 to 31 across $Re = 10 - 2000$) with no manual tuning. This comparison is preliminary: POD and adaptive-SVD baselines are Phase 2 work.

6. Honest Ablation: CFL Steering Is Not Claimed

Associator-adaptive time stepping shows roughly 10% lower L2 error at $Re \geq 100$. We ran the control experiment a reviewer would run: a classical solver with a uniform CFL reduction ($cfl = 0.362$, no octonions) reproduces the same accuracy at $Re \geq 500$ at roughly 60x lower runtime. The steering gain survives the control only at $Re = 100$.

We therefore do not claim a net steering advantage on this benchmark. The associator's measured contribution is rank selection (Section 5). Whether adaptive steering wins on spatially heterogeneous flows - where a constant factor cannot adapt - is a Phase 2 question.

7. Quantum Hardware Pathway (No Hardware Claimed)

The Tucker decomposition maps directly to a tree tensor network (TTN) state. Under the standard $\log_2$ encoding, each mode of dimension n requires $\text{ceil}(\log_2 n)$ qubits. Using the measured saturated ranks:

Grid	Logical qubits	Physical qubits (surface code)	Basis
64 ²	~46	~46,000	measured ranks
128 ²	~50	~50,000	measured ranks
256 ²	~54	~54,000	measured ranks
512 ²	~57	~57,000	projected

Qubit count grows logarithmically with grid size while classical storage grows linearly - the credible path to exponential quantum memory advantage under fault-tolerant hardware. Current devices (≤ 133 qubits, shallow coherent depth) cannot execute the required $O(10^4)$ -gate preparation circuits; this is a mapping, not a result.

8. What Is Explicitly Not Claimed

- No quantum-hardware speedup of any kind.
- No net CFL-steering advantage (our control experiment partially falsified it; reported as an ablation).
- No wall-clock reduced-order-model speedup (operation-count estimate only; DEIM hyper-reduction is Phase 2).
- No claim that TGV compression ratios transfer unchanged to industrial flows (the perturbed-flow study measures the degradation direction: graceful).

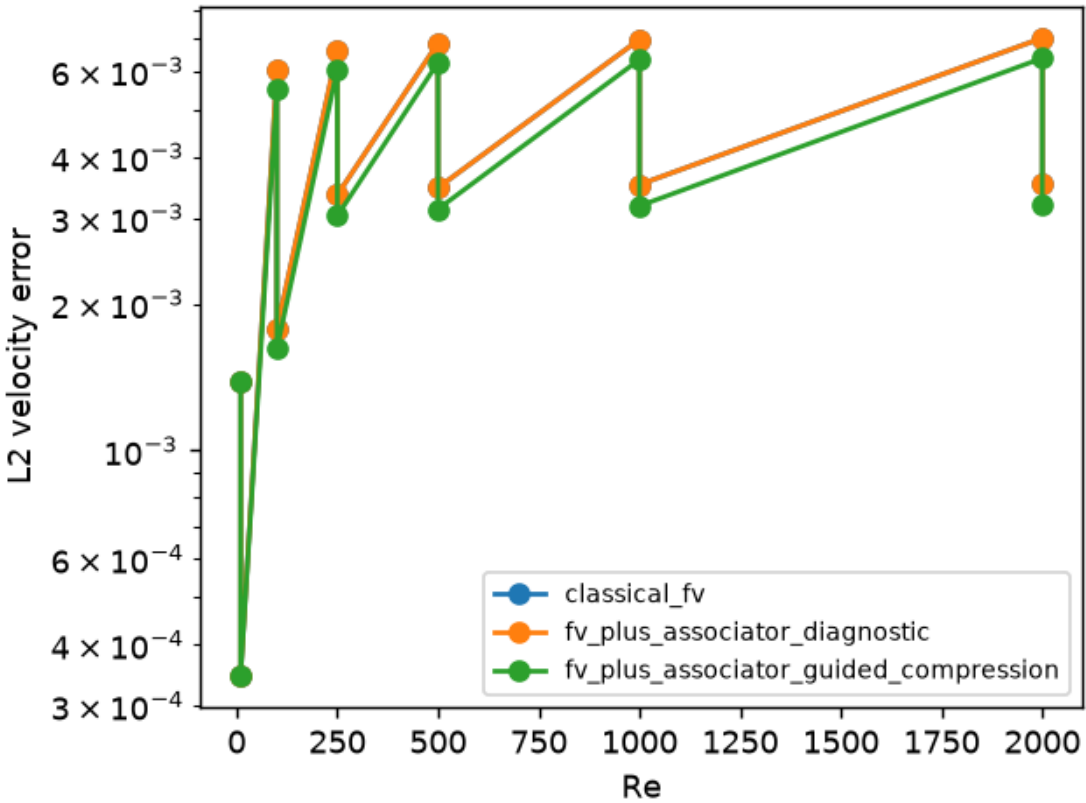
9. Reproducibility

All results regenerate on a standard Windows CPU with NumPy, TensorLy, pandas, matplotlib and pytest. No GPU, cluster, or external CFD data is required. The scripts that produced each reported number:

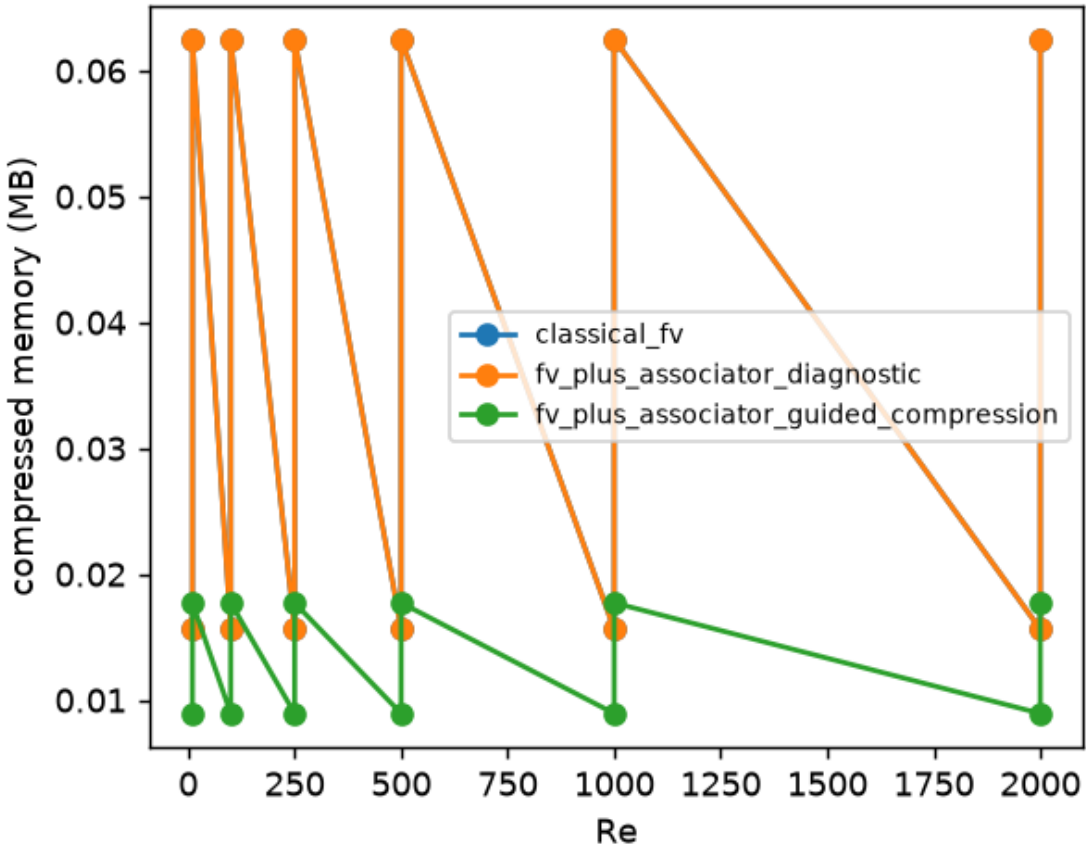
```
python -m pytest # 48/48 tests
python scripts/run_reynolds_sweep.py --config configs/reynolds_sweep.yaml
python scripts/run_advantage_study.py # 3-way comparison
python scripts/run_scalability_projection.py --re 100 --grids 32 64 128 256
python scripts/run_perturbed_advantage_study.py --re 100 --nx 64
python scripts/run_trajectory_compression.py --re 100 --nx 64 --t-final 0.5
```

The accompanying uploads contain the raw metrics (TGV_Scaling_Metrics.csv), the error-scaling figure (L2_Error_vs_Reynolds_Comparison.png), the finite-volume solver source (finite_volume_solver.py), and the tensor-network compression source (tensor_network_compression.py).

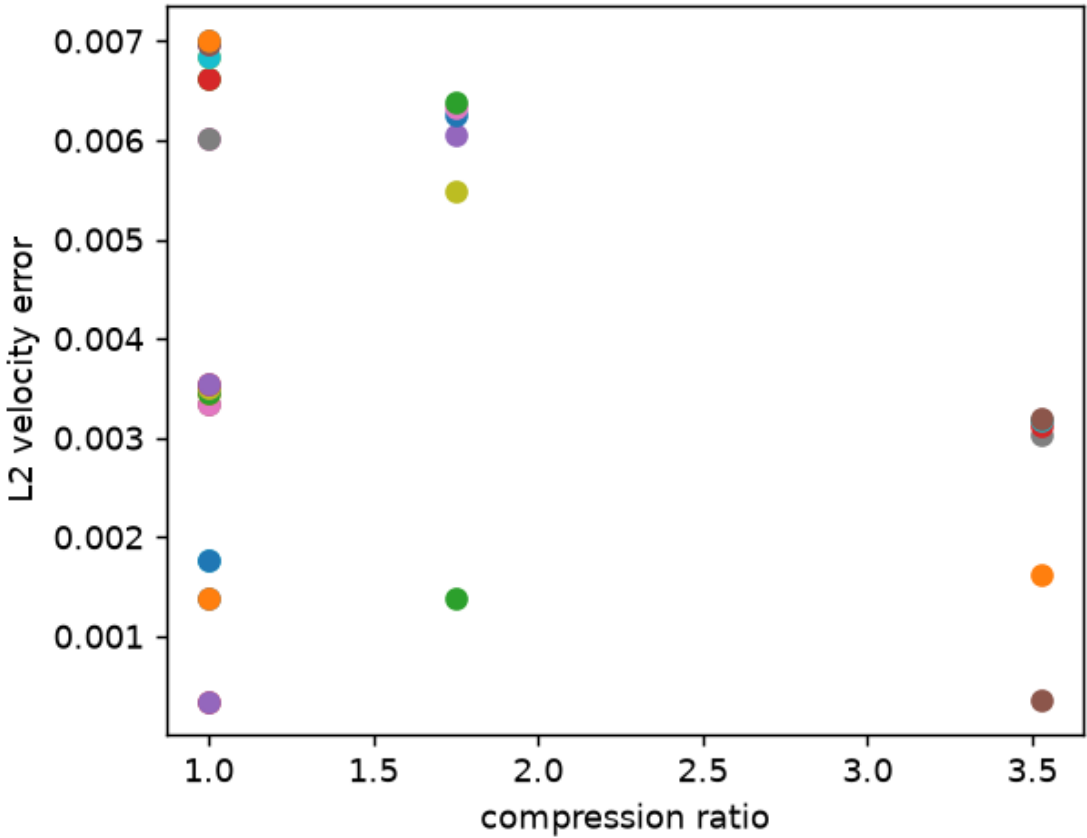
10. Key Figures



L2 velocity error vs Reynolds number, all methods (32^2 and 64^2 grids).



Memory footprint vs Reynolds number: dense classical vs compressed storage.



Compression ratio vs reconstruction error trade-off.

